

Total marks – 120.

Attempt Questions 1 – 8.

All questions are of equal value.

Answer each question on a NEW page. Extra paper is available.

Question 1 (15 marks) Use a NEW sheet of paper,

a) $\int x e^{x^2} dx$ 1

b) Evaluate $\int_1^3 x^2 \log_e x dx$ 3

c) $\int \frac{dx}{5 + 4 \cos x}$ 3

d) True or false:

i) $\int_{-1}^1 \sin^7 x dx = 0$ 1

ii) $\int_{-\pi}^{\pi} x \cos x dx = 2 \int_0^{\pi} x \cos x dx$ 1

iii) $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ 1

e) i) Write $\frac{x^2 + 2x + 3}{(x+1)(x^2+1)}$ in the form $\frac{A}{x+1} + \frac{Bx+C}{x^2+1}$. 2

ii) Hence find $\int \frac{x^2 + 2x + 3}{(x+1)(x^2+1)} dx$ 3

Question 4 (15 marks) Use a NEW sheet of paper

a) The polynomial $x^3 - 4x^2 + x = 0$ has roots α, β and γ . Find a polynomial with integer coefficients that has roots α^2, β^2 and γ^2 . 3

b) Use the method of cylindrical shells to find the volume of the solid generated when the area between $y = \sqrt{1-x^2}$ and $y = x$ from $x = 0$ to $x = \frac{1}{\sqrt{2}}$ is rotated about the y -axis. 3

c) i) Let n be a positive integer and $I_n = \int_1^2 (\log_e x)^n dx$.

Show that $I_n = 2(\log_e 2)^n - nI_{n-1}$. 3

ii) Hence evaluate $\int_1^2 (\log_e x)^3 dx$. 2

d) $P(x)$ is a polynomial where $P(x) = A(x)(x-k)^n$ where $A(x)$ is a polynomial and n is an integer greater than 1.

i) Show that $P(k) = P'(k) = 0$. 2

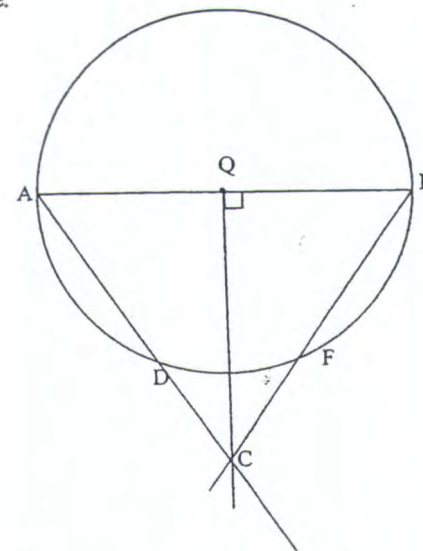
ii) Hence or otherwise find the values of u and v if $x^4 + ux^3 + 9x^2 + vx + 2 = 0$ has a triple root at $x = 3$. 2

Question 5 (15 marks) Use a NEW sheet of paper

a) AB is the diameter of the circle, centre Q. CQ is perpendicular to AB. AC meets the circle at D, BC meets the circle at F.

Show that CDQB is a cyclic quadrilateral. 3

Not to scale.



Question 5 is continued on the next page

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Question 2 (15 marks) Use a NEW sheet of paper.

a) If $z = 2 + 3i$ find in $x + iy$ form

i) $2z + 3\bar{z}$

ii) $\frac{1}{z}$

iii) z^2

b) i) Write in $(1 + i)$ in modulus-argument form.

ii) Hence determine $(1 + i)^{10}$. Write your answer in $x + iy$ form.

c) Sketch on an Argand diagram the complex numbers that satisfy both

$$|z - 2 + i| < 3 \text{ and } -\frac{\pi}{4} \leq \arg(z) \leq \frac{\pi}{4}$$

d) If $z_2 = iz_1$ show that $z_2 - z_1 = i(z_1 + z_2)$. Interpret this geometrically.

e) The complex number z is a function of the real number t where

$$z = \frac{t-i}{t+i}, 0 \leq t \leq 1. \text{ Evaluate } |z| \text{ and hence describe the locus of } z.$$

Question 3 (15 marks) Use a NEW sheet of paper.

(a) i) Sketch $f(x) = x^3 - 3x^2$ showing x and y intercepts and all stationary points.

ii) Hence sketch:

$\alpha) y = (f(x))^2$

$\beta) y = \frac{1}{f(x)}$

$\gamma) y = \sqrt{f(x)}$

$\delta) y = f(|x|)$

b) Draw a clear sketch of $\frac{x^2}{4} - \frac{y^2}{9} = 1$ showing asymptotes, foci and directrices.

c) Find the gradient of the tangent to $xy + 2x = 4$ at $(1, 2)$.

Question 6 (cont)

- b) Using Mathematical induction, show that for each positive integer n , there are unique positive integers p_n and q_n such that $(1 + \sqrt{2})^n = p_n + q_n\sqrt{2}$ and that $p_n^2 - 2q_n^2 = (-1)^n$. 5

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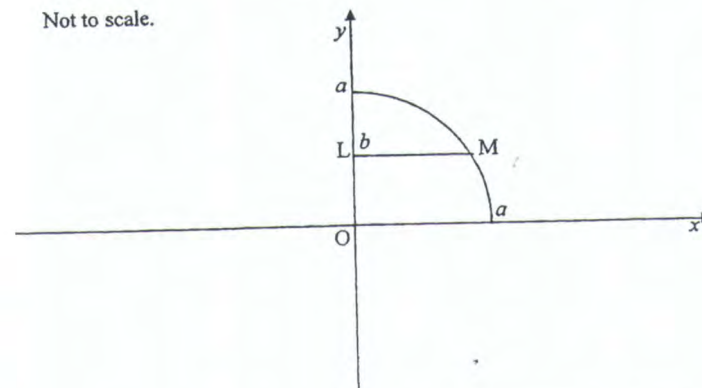
- c) A particle of mass m is projected vertically upwards with initial velocity u , in a medium whose resistance to motion varies as the velocity of the particle ie $\ddot{x} = -g - kv$.

- i) Show that the time taken to reach the highest point is $\frac{1}{k} \log_e \left(1 + \frac{ku}{g}\right)$. 3
- ii) Find the greatest height the particle will reach. 3

Question 7 (15 marks) Use a NEW sheet of paper

- a) The horizontal interval LM through the point $(0, b)$, where $0 < b < a$, divides the area between the curve $x^2 + y^2 = a^2$ and the coordinate axes into 2 equal parts.

Not to scale.



- i) By finding the area between LM, the coordinate axes and the circle,

$$\text{show that } \sin^{-1} \frac{b}{a} + \frac{b\sqrt{a^2 - b^2}}{a^2} = \frac{\pi}{4}.$$

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- ii) If the radius of the circle is 1 unit, show that b can be found by solving

$$\sin 2\theta = \frac{\pi}{2} - 2\theta, \text{ where } \theta = \sin^{-1} b.$$

2

- iii) Describe how θ and hence b could be approximated.

1

Question 7 is continued on the next page

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Question 5 (cont)

b) If $z = \cos \theta + i \sin \theta$:

i) Show that $z^n + z^{-n} = 2 \cos n\theta$ where n is a positive integer. 1

ii) Given that $(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$ find an expression for

$(z + \frac{1}{z})^4$ in the form $a \cos 4\theta + b \cos 2\theta + c$. 3

iii) Hence evaluate $\int_0^{\pi} \cos^4 \theta d\theta$. 2

c) $P(a \sec \theta, b \tan \theta)$ is a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. $S(ae, 0)$ is a focus with corresponding directrix $x = \frac{a}{e}$.

i) Using the focus-directrix definition of an hyperbola or otherwise, prove that $PS = a(e \sec \theta - 1)$. 2

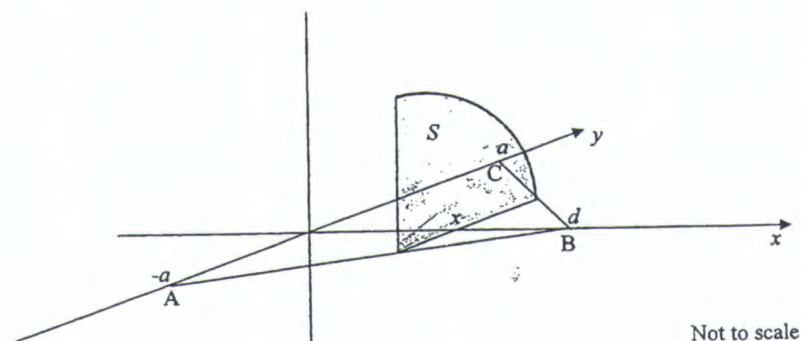
ii) Show that the normal at P has the equation

$$y - b \tan \theta = \frac{-a \tan \theta}{b \sec \theta} (x - a \sec \theta).$$
2

iii) The normal at P meets the x -axis at G . Show that $\frac{SG}{SP} = e$. 2

Question 6 (15 marks) Use a NEW sheet of paper

a) The base of a solid is the isosceles triangle $\triangle ABC$. The triangle has perpendicular height d and has a base $2a$. Each cross section perpendicular to the x -axis is a quarter circle with the radius in the x - y plane and parallel to the y -axis. A typical cross section S is shown positioned at x on the x -axis.



i) Show that the area of S is $\frac{\pi a^2}{d^2} (d - x)^2$. 2

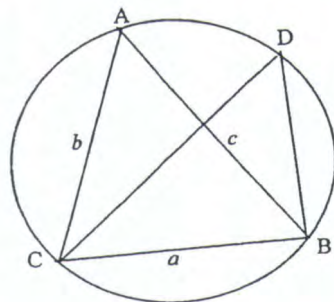
ii) Hence find the volume of the solid. 2

Question 6 is continued on the next page

Question 8 (cont)

b) ABC is a triangle inscribed in a circle of radius r . The centre of the circle is in the interior of the triangle. Let $a = BC$, $b = AC$ and $c = AB$.

Not to scale.



i) Let CD be a diameter. Use the sine rule to show that $\frac{a}{\sin A} = 2r$. 2

ii) Show that the area of the triangle ABC = $\frac{1}{2}r^2 (\sin 2A + \sin 2B + \sin 2C)$. 2

iii) Show that for angles A, B and C in a triangle, 4

$$\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$$

(iv) Hence show that the area of triangle ABC is $\frac{abc}{4r}$ 2

END OF EXAMINATION

Question 1

$$(a) \int x e^{x^2} dx = \frac{1}{2} \int 2x e^{x^2} dx = \frac{1}{2} e^{x^2} + C \quad (\text{or Sub } u = x^2)$$

$$(b) \text{ By parts } u = \ln x \quad v' = x^2 \\ \int_1^3 x^2 \ln x dx = \left[\frac{1}{3} x^3 \ln x \right]_1^3 - \int_1^3 \frac{1}{3} x^3 \cdot \frac{1}{x} dx \\ = \frac{1}{3} 3^3 \ln 3 - 0 - \frac{1}{3} \int_1^3 x^2 dx \\ = 9 \ln 3 - \frac{1}{3} \left[\frac{x^3}{3} \right]_1^3 = 9 \ln 3 - \frac{26}{9}$$

$$(c) \text{ Let } t = \tan \frac{x}{2} \\ dt = \frac{1}{2} \sec^2 \frac{x}{2} dx = \frac{1}{2} (1+t^2) dx \\ \therefore dx = \frac{2 dt}{1+t^2} \\ \int \frac{dx}{5+4\cos x} = \int \frac{1}{5 + \frac{4(1-t^2)}{1+t^2}} \cdot \frac{2 dt}{1+t^2} \\ = 2 \int \frac{dt}{5(1+t^2) + 4(1-t^2)} = 2 \int \frac{dt}{9+t^2} \\ = \frac{2}{3} \tan^{-1} \left(\frac{t}{3} \right) = \frac{2}{3} \tan^{-1} \left(\frac{1}{3} \tan \frac{x}{2} \right) + C$$

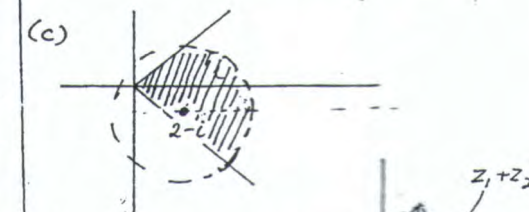
(d) (i) TRUE (odd function)
(ii) FALSE (odd function)
(iii) TRUE (standard result)

$$(e) (i) x^2 + 2x + 3 = A(x+1) + (Bx+C)(x+1) \\ \text{let } x = -1 \quad 2A = 2 \quad \therefore A = 1 \\ \text{Equate } x^2 \quad 1 = A + B \quad \therefore B = 0 \\ \text{Equate constant } 3 = A + C \quad \therefore C = 2 \\ \int \frac{x^2 + 2x + 3}{(x+1)(x^2+1)} dx = \frac{1}{x+1} + \frac{2}{x^2+1} \\ = \ln|x+1| + 2 \tan^{-1} x + C$$

Question 2 (a)

$$(i) 10 - 3i \\ (ii) \frac{1}{2+3i} \cdot \frac{2-3i}{2-3i} = \frac{2-3i}{13} \\ (iii) (2+3i)^2 = 4 + 12i - 9 = -5 + 12i$$

$$(b) (i) |1+i| = \sqrt{2} \\ \arg(1+i) = \pi/4 \\ \therefore 1+i = \sqrt{2} \operatorname{cis} \frac{\pi}{4} \\ (ii) (1+i)^{10} = (\sqrt{2})^{10} \operatorname{cis} \frac{10\pi}{4} = 2^5 \operatorname{cis} \frac{\pi}{2} = 32i$$



(d) z_2 is a rotation of z_1 by $\pi/2$. This creates a square. $z_2 - z_1$ and $z_1 + z_2$ are diagonals. If $z_2 - z_1 = i(z_1 + z_2)$ Then this states that the diagonals are perpendicular. Take modulus of each side $|z_2 - z_1| = |z_1 + z_2|$ This states diagonals are equal.

$$\text{OR algebraically} \\ z_2 - z_1 = iz_1 - z_1 = iz_1 + iz_1 = i(z_1 + iz_1)$$

Geometrical interpretation as above.

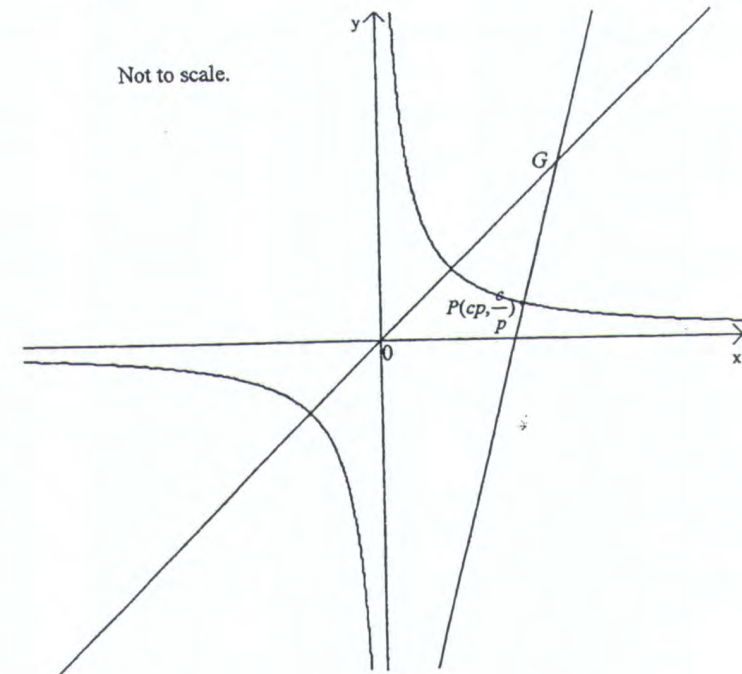
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Question 7 (cont)

- b)
- Show that $y^2 = (1-x)^2(4-x)$ has 2 stationary points at $x = 3$ and determine their nature. 3
 - Draw a labelled sketch of $y^2 = (1-x)^2(4-x)$ indicating any relevant points. 2
- c) Ten cards are numbered from 1 to 10. They are dealt randomly to 9 people, with eight people receiving 1 card and 1 person receiving 2 cards.
- What is the probability that the person receiving 2 cards is dealt two odd cards? 2
 - What is the probability that this person is dealt one odd card and even card? 1

Question 8 (15 marks) Use a NEW sheet of paper

- a) The normal at $P(cp, \frac{c}{p})$, where $p^2 \neq 1$, on the hyperbola $xy = c^2$ meets $y = x$ at G .



- Show that the equation of the normal at P is $p^3x - py = c(p^4 - 1)$. 1
- Find the length PG in terms of p . 2
- Show that the distance PG is greater than $c\sqrt{2}$. 2

Question 8 is continued on the next page

6b) $z^n + z^{-n} = \cos n\theta + i \sin n\theta + \cos(-n\theta) + i \sin(-n\theta)$
by de Moivre

$$= \cos n\theta + \cos n\theta + i \sin n\theta - i \sin n\theta$$

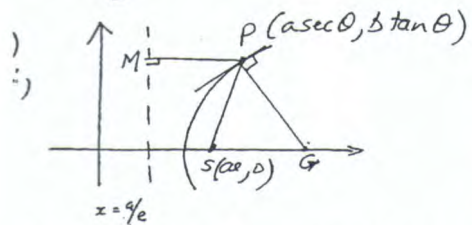
$$= 2 \cos n\theta$$

ii) $(z + \frac{1}{z})^4 = z^4 + 4z^2 + 6 + 4z^{-2} + z^{-4}$
 $= (z^4 + z^{-4}) + 4(z^2 + z^{-2}) + 6$
 $= 2 \cos 4\theta + 8 \cos 2\theta + 6$

Also $(z + \frac{1}{z})^4 = (2 \cos \theta)^4$
 $= 16 \cos^4 \theta$

$$\therefore \cos^4 \theta = \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8}$$

i) $\int_0^{\pi} \cos^4 \theta d\theta$
 $= \left[\frac{1}{32} \sin 4\theta + \frac{1}{4} \sin 2\theta + \frac{3\theta}{8} \right]_0^{\pi}$
 $= \frac{3\pi}{8}$



from diagram & definition

$$PS = e \cdot PM$$

$$= e[a \sec \theta - a]$$

$$= a[e \sec \theta - 1]$$

i) $\frac{2x}{a^2} - \frac{2y y'}{b^2} = 0$ (differentiating)
 $\therefore y' = \frac{b^2 x}{a^2 y}$

gradient at P = $\frac{b^2 \sec \theta}{a^2 b \tan \theta}$
 $= \frac{b \sec \theta}{a \tan \theta}$

$\therefore$ gradient of normal = $-\frac{a \tan \theta}{b \sec \theta}$

Equation of normal

$$y - b \tan \theta = -\frac{a \tan \theta}{b \sec \theta} (x - a \sec \theta)$$

(iii) G is where normal meets x-axis, i.e. $y=0$

$$\therefore -b \tan \theta = -\frac{a \tan \theta}{b \sec \theta} (x - a \sec \theta)$$

$$\therefore \frac{a}{b \sec \theta} (x - a \sec \theta) = b$$

$$a(x - a \sec \theta) = b^2 \sec \theta$$

$$\therefore ax = (a^2 + b^2) \sec \theta$$

$$= (a^2 + a^2(e^2 - 1)) \sec \theta$$

$$\therefore x = a e^2 \sec \theta$$

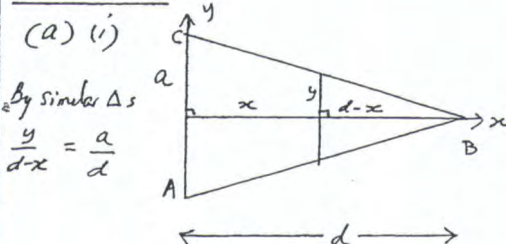
$$\therefore SG = a e^2 \sec \theta - a e$$

$$= a e (e \sec \theta - 1)$$

$$\therefore \frac{SG}{SP} = \frac{a e (e \sec \theta - 1)}{a (e \sec \theta - 1)}$$

$$= e$$

Question 6.



$\therefore y = \frac{a}{d}(d-x)$

gradient of line BC = $-\frac{a}{d}$
y-intercept = a
Equation of BC = $y = -\frac{a}{d}x + a$
 $\therefore y = \frac{a}{d}(d-x)$

Area S = $\frac{1}{2} \pi (2y)^2 = \pi y^2$
 $= \frac{\pi a^2}{d^2} (d-x)^2$

(ii) Vol = $\int_0^d \frac{\pi a^2}{d^2} (d-x)^2 dx$
 $= \frac{\pi a^2}{d^2} \left[-\frac{(d-x)^3}{3} \right]_0^d$
 $= \frac{\pi a^2}{d^2} \left[0 - -\frac{d^3}{3} \right] = \frac{\pi a^2 d}{3}$

(b) When $n=1$
 $(1+\sqrt{2})^1 = 1+\sqrt{2} \therefore p_1=1 \text{ and } q_1=1$
 $(p_1)^2 - 2(q_1)^2 = 1-2 = (-1)^1$
 $\therefore$ True when $n=1$.

Assume true for $n=k$
 $(1+\sqrt{2})^{k+1} = (1+\sqrt{2})^k (1+\sqrt{2})$
 $= (p_k + q_k \sqrt{2})(1+\sqrt{2})$
by assumption
 $= p_k + p_k \sqrt{2} + q_k \sqrt{2} + 2q_k$
 $= (p_k + 2q_k) + (p_k + q_k) \sqrt{2}$

Hence $p_{k+1} = p_k + 2q_k$ and
 $q_{k+1} = p_k + q_k$
(These are uniquely determined since p_k and q_k are)
 $p_{k+1}^2 - 2q_{k+1}^2 =$
 $= (p_k + 2q_k)^2 - 2(p_k + q_k)^2$

$$= p_k^2 + 4p_k q_k + 4q_k^2 - 2p_k^2 - 4p_k q_k - 2q_k^2$$

$$= -p_k^2 + 2q_k^2 = -(p_k^2 - 2q_k^2)$$

$$= -(-1)^k \text{ by hypothesis}$$

$$= (-1)^{k+1}$$

(c) (i) $\frac{dv}{dt} = -g - kv$
 $\frac{dt}{dv} = -\frac{1}{g+kv} \therefore t = -\int \frac{1}{g+kv} dv$
 $= -\frac{1}{k} \ln(g+kv) + C$
When $t=0$ $v=u$
 $0 = -\frac{1}{k} \ln(g+ku) + C$
 $\therefore t = \frac{1}{k} \ln(g+ku) - \frac{1}{k} \ln(g+kv)$
 $= \frac{1}{k} \ln\left(\frac{g+ku}{g+kv}\right)$
When $v=0$ (for max. height)
 $t = \frac{1}{k} \ln\left(\frac{g+ku}{g}\right) = \frac{1}{k} \ln\left(1 + \frac{ku}{g}\right)$

(ii) Easier to start from equation of motion
 $v \frac{dv}{dx} = -g - kv = -(g+kv)$
 $\frac{dv}{dx} = -\left(\frac{g+kv}{v}\right) \frac{dx}{dv} = -\frac{v}{g+kv}$
 $\therefore x = -\int \frac{v}{g+kv} dv = -\frac{1}{k} \int \frac{g+kv-g}{g+kv} dv$
 $= -\frac{1}{k} \int \left(1 - \frac{g}{g+kv}\right) dv$
 $= -\frac{1}{k} \left[v - \frac{g}{k} \ln(g+kv) \right] + C$

When $x=0$ $v=u$
 $\therefore 0 = -\frac{1}{k} \left[u - \frac{g}{k} \ln(g+ku) \right] + C$
 $\therefore x = \left(\frac{u}{k} - \frac{v}{k}\right) + \frac{g}{k^2} \ln\left(\frac{g+kv}{g+ku}\right)$
for max height let $v=0$
 $\therefore x = \frac{u}{k} + \frac{g}{k^2} \ln\left(\frac{g}{g+ku}\right)$
OR $\frac{u}{k} - \frac{g}{k^2} \ln\left(1 + \frac{ku}{g}\right)$

$1+t+2i \quad \sqrt{x+1}$
 since t is real
 when $t=0 \quad z = \frac{-i}{x} = -1$
 when $t=1 \quad z = \frac{1-i}{1+i} \cdot \frac{1-i}{1-i}$
 $= \frac{1-2i+1}{2} = -i$

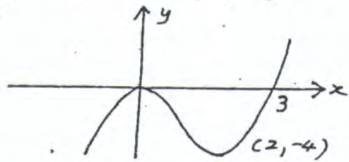
Locus is $1/4$ circle



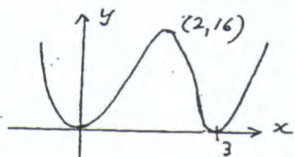
Question 3.

(a) (i) $y = x^3 - 3x^2 \quad y' = 3x^2 - 6x$
 $y'' = 6x - 6$

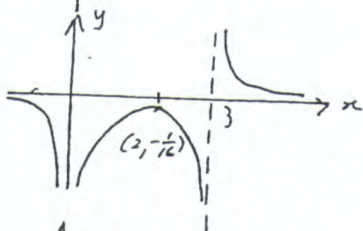
St. No $y' = 0 \Rightarrow x = 0$ or 2
 $y''(0) = -6 \therefore (0,0)$ is MAX
 $y''(2) = 6 \therefore (2,-4)$ is MIN



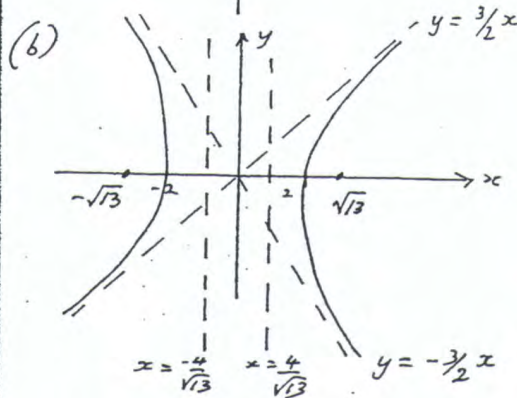
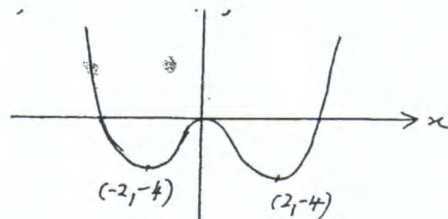
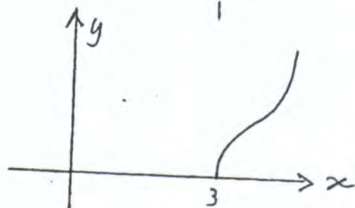
(ii) (2)



(3)



(r)

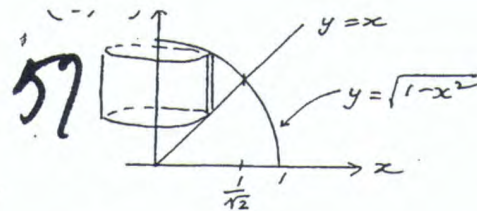


$a=2 \quad b=3$
 $b^2 = a^2(e^2 - 1) \therefore 9 = 4(e^2 - 1)$
 $\therefore e^2 - 1 = 9/4 \quad e^2 = 13/4 \quad e = \sqrt{13}/2$
 $ae = \sqrt{13} \quad \frac{a}{e} = \frac{4}{\sqrt{13}}$

(c) $xy' + y + 2 = 0$
 $y' = -\left(\frac{y+2}{x}\right)$
 At $(1,2) \quad y' = -\left(\frac{2+2}{1}\right) = -4$

Question 4

(a) Replace x by $\sqrt{x}$
 $(\sqrt{x})^3 - 4x + \sqrt{x} = 0$
 $x\sqrt{x} - 4x + \sqrt{x} = 0$
 $\sqrt{x}(x+1) = 4x$
 Square both sides
 $x(x^2 + 2x + 1) = 16x^2$
 $x^3 - 14x^2 + x = 0$



$\delta v = 2\pi r h \delta x$
 $= 2\pi x [\sqrt{1-x^2} - x] \delta x$
 $\text{Vol} = \lim \sum \delta v$
 $= 2\pi \int_{-1/\sqrt{2}}^{1/\sqrt{2}} (x\sqrt{1-x^2} - x^2) dx$
 $= 2\pi \left[-\frac{1}{2} \cdot \frac{2}{3} (1-x^2)^{3/2} - \frac{x^3}{3} \right]_{-1/\sqrt{2}}^{1/\sqrt{2}}$
 $= \pi \left[-\frac{1}{3} \cdot \frac{1}{2\sqrt{2}} - \frac{1}{6\sqrt{2}} - \left(-\frac{1}{3} - 0\right) \right]$
 $= 2\pi \left[\frac{1}{3} - \frac{1}{3\sqrt{2}} \right]$

(c) (i) $I_n = \int (\log_e x)^n dx$
 [By parts $u = (\log_e x)^n \quad v' = 1$]
 $\therefore I_n = [x(\log_e x)^n]_1^2 - \int_1^2 x \cdot n(\log_e x)^{n-1} \cdot \frac{1}{x} dx$

$= 2(\log_e 2)^n - n \int_1^2 (\log_e x)^{n-1} dx$
 $= 2(\log_e 2)^n - n I_{n-1}$
 (ii) $I_0 = \int_1^2 dx = 1$
 $I_1 = 2 \log_e 2 - 1$
 $I_2 = 2(\log_e 2)^2 - 2[2 \log_e 2 - 1]$
 $= 2(\log_e 2)^2 - 4 \log_e 2 + 2$
 $I_3 = 2(\log_e 2)^3 - 3[2(\log_e 2)^2 - 4 \log_e 2 + 2]$
 $= 2(\log_e 2)^3 - 6(\log_e 2)^2 + 12 \log_e 2 - 6$

(d) (i) $P(x) = A(x)(x-k)$
 $P'(x) = A(x)n(x-k)^{n-1} + A'(x)(x-k)^n$
 $= (x-k)^{n-1} [nA(x) + A'(x)(x-k)]$
 $P(k) = A(k)(k-k)^n = 0$
 $P'(k) = (k-k)^{n-1} [nA(k) + A'(k)(k-k)] = 0$

(ii) Triple root at $x=3$
 $\therefore P(3) = P'(3) = P''(3) = 0$
 $P(x) = x^4 + ux^3 + 9x^2 + vx + w$
 $P'(x) = 4x^3 + 3ux^2 + 18x + v$
 $P''(x) = 12x^2 + 6ux + 18$
 $P''(3) = 0 \Rightarrow 108 + 18u + 18 = 0$
 $\therefore 18u = -126$
 $\therefore u = -7$
 $P'(3) = 0 \Rightarrow 108 - 189 + 54 + v = 0$
 $\therefore v = 27$

Question 5

(a) $\angle ADB = 90^\circ$ ($\angle$ in semicircle)
 $\angle AQC = 90^\circ$ (given)
 $\therefore CDQB$ is cyclic
 (Converse of $\angle$'s in same seg)

OTHER WAYS ARE POSSIBLE
 We can prove $\triangle AQC \cong \triangle BQC$
 Let $\angle QAC = \theta$, then $\angle QBC = \theta$
 (corr sides congruent)
 Also since $\triangle AQC$ is isosceles
 $\angle AQC = \theta$
 Hence $\angle AQC = \angle BQC = \theta$
 $\therefore CDQB$ is cyclic
 (exterior $\angle =$ opp. interior $\angle$)

(iii) Since A, B, C are angles in a triangle
 $C = \pi - (A+B)$

$$\begin{aligned} \therefore \sin 2A + \sin 2B + \sin 2C \\ &= \sin 2A + \sin 2B + \sin 2(\pi - (A+B)) \\ &= \sin 2A + \sin 2B - \sin (2A + 2B) \\ &\quad (\sin (2\pi - \theta) = -\sin \theta) \\ &= \sin 2A + \sin 2B - \sin 2A \cos 2B - \cos 2A \sin 2B \\ &= \sin 2A (1 - \cos 2B) + \sin 2B (1 - \cos 2A) \\ &= \sin 2A \cdot 2 \sin^2 B + \sin 2B \sin^2 A \\ &\quad (\cos 2\theta = 1 - 2\sin^2 \theta) \\ &= 4 \sin A \cos A \sin^2 B + 4 \sin B \cos B \sin^2 A \\ &= 4 \sin A \sin B (\cos A \sin B + \sin A \cos B) \\ &= 4 \sin A \sin B \sin (A+B) \\ &= 4 \sin A \sin B \sin (\pi - C) \\ &= 4 \sin A \sin B \sin C \end{aligned}$$

(iv) Area $\triangle ABC$

$$= \frac{1}{2} r^2 (\sin 2A + \sin 2B + \sin 2C)$$

from (ii)

$$= \frac{1}{2} r^2 \cdot 4 \sin A \sin B \sin C$$

from (iii)

From (i) $\sin A = \frac{a}{2r}$

$$\sin B = \frac{b}{2r} \quad \sin C = \frac{c}{2r}$$

$$\begin{aligned} \therefore \text{Area } \triangle ABC &= \frac{1}{2} r^2 \cdot 4 \cdot \frac{a}{2r} \cdot \frac{b}{2r} \cdot \frac{c}{2r} \\ &= \frac{abc}{4r} \end{aligned}$$

OR

$$\begin{aligned} \text{Area} &= \frac{1}{2} bc \sin A \\ &= \frac{1}{2} bc \cdot \frac{a}{2r} \end{aligned}$$

from (i)

$$= \frac{abc}{4r}$$

It is possible to prove (iii) from (ii) and (iv) since (iv) can be proven independently from (iii) :

From (iv)

$$\text{Area } \triangle ABC = \frac{abc}{4r}$$

From (i)

$$\begin{aligned} a &= 2r \sin A \\ b &= 2r \sin B \\ c &= 2r \sin C \end{aligned}$$

$$\begin{aligned} \therefore \text{Area } \triangle ABC &= \frac{(2r)^3 \sin A \sin B \sin C}{4r} \\ &= 2r^2 \sin A \sin B \sin C \end{aligned}$$

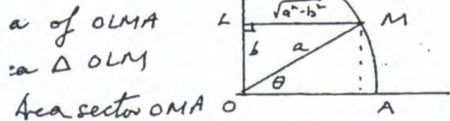
From (ii)

$$\text{Area } \triangle ABC = \frac{1}{2} r^2 (\sin 2A + \sin 2B + \sin 2C)$$

Equating those gives

$$\begin{aligned} \sin 2A + \sin 2B + \sin 2C \\ &= 4 \sin A \sin B \sin C \end{aligned}$$

sum 1.



a of OLMA
 ΔOLM
 Area sector OMA
 $\frac{1}{2} OL \cdot LM + \frac{1}{2} a^2 \theta$
 $b\sqrt{a^2-b^2} + \frac{1}{2} a^2 \sin^{-1}\left(\frac{b}{a}\right)$

OR
 area OLMA = $\int_0^b \sqrt{a^2-y^2} dy$
 angle variable: Let $y = a \sin \theta$

$\therefore \frac{dy}{d\theta} = a \cos \theta$
 $\int_0^b \sqrt{a^2-y^2} dy = \int_0^{\sin^{-1}(b/a)} \sqrt{a^2-a^2 \sin^2 \theta} \cdot a \cos \theta d\theta$

$\int_0^{\sin^{-1}(b/a)} \cos^2 \theta d\theta = \frac{a^2}{2} \int_0^{\sin^{-1}(b/a)} (\cos 2\theta + 1) d\theta$

$\frac{a^2}{2} \left[\frac{1}{2} \sin 2\theta + \theta \right]_0^{\sin^{-1}(b/a)}$
 $\frac{a^2}{2} \left[\sin \theta \cos \theta + \theta \right]_0^{\sin^{-1}(b/a)}$

$\left[\frac{b\sqrt{a^2-b^2}}{a} + \sin^{-1}\left(\frac{b}{a}\right) \right] \cdot \frac{a^2}{2}$
 $\sin \theta = \frac{b}{a}$

$\frac{1}{2} b\sqrt{a^2-b^2} + \frac{1}{2} a^2 \sin^{-1}\left(\frac{b}{a}\right)$
 $\therefore \cos \theta = \frac{\sqrt{a^2-b^2}}{a}$

area OLMA = $\frac{1}{2}$ area of $\frac{1}{4}$ circle
 $= \frac{\pi a^2}{8}$

adding + multiplying by $\frac{2}{a^2}$:

$\sqrt{a^2-b^2} + \sin^{-1}\left(\frac{b}{a}\right) = \frac{\pi}{4}$

(i) Lemny
 $b\sqrt{1-b^2} + \sin^{-1} b = \frac{\pi}{4}$
 if $\sin^{-1} b = \theta$
 then $\sin \theta = b$
 $\cos \theta = \sqrt{1-b^2}$

$\therefore \sin \theta \cos \theta + \theta = \frac{\pi}{4}$
 $\therefore 2 \sin \theta \cos \theta + 2\theta = \frac{\pi}{2}$
 $\therefore \sin 2\theta = \frac{\pi}{2} - 2\theta$

(iii) Various answers: e.g.
 graphically or Newton's Method

(b) $y^2 = (1-x)(4-x)$
 $\therefore 2yy' = (1-x)(-1) + (4-x)(-1)$
 $= (1-x)(3x-9)$

$\therefore y' = \frac{3(1-x)(x-3)}{y}$

[Note y' is undefined at $x=1$]

When $x=3$ $y'=0$

when $x=3$ $y^2=4$

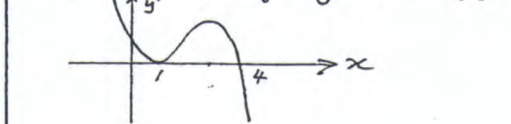
$\therefore y = \pm 2$

When $y=2$ $\frac{x}{y'} \mid \begin{matrix} 2 & 3 & 4 \\ 1 & 0 & - \end{matrix}$ MAX

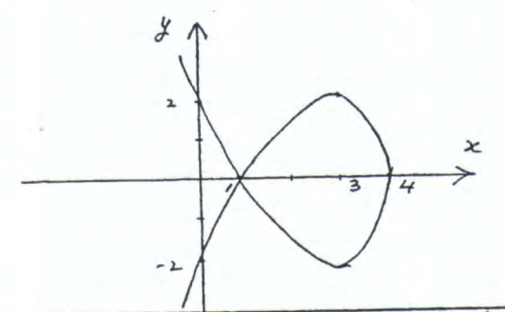
When $y=-2$ $\frac{x}{y'} \mid \begin{matrix} 2 & 3 & 4 \\ -1 & 0 & + \end{matrix}$ MIN

$\therefore$ Two St. pts
 $(3, 2)$ is a max.
 $(3, -2)$ is a min.

Here is the graph of $y = (1-x)^2(4-x)$



$\Rightarrow y = \pm \sqrt{(1-x)^2(4-x)}$ 38



(c) It is irrelevant who gets the 2 cards

(i) # ways someone can be dealt 2 cards = ${}^{10}C_2$

ways someone can be dealt 2 odd cards = 5C_2

$\therefore \text{Prob} = \frac{{}^5C_2}{{}^{10}C_2} = \frac{2}{9}$

(ii) Prob (2 even cards) = Prob (2 odd)

$\therefore \text{Prob (1 of each)} = 1 - 2 \times \frac{2}{9} = \frac{5}{9}$

Question 8:

(a) (i) $xy = c^2 \therefore xy' + y = 0$

$\therefore y' = -\frac{y}{x} \therefore m_{\perp} = \frac{x}{y}$

At P $m_{\perp} = \frac{cp}{cp} = p^2$

Equation of normal:

$y - \frac{c}{p} = p^2(x - cp)$

or $p^3x - py = c(p^4 - 1)$

on rearranging.

(ii) At G $y=x$

$\therefore p^3x - px = c(p^4 - 1)$

$x(p^2 - 1)p = c(p^2 - 1)(p^2 + 1)$

$p^2 \neq 1 \therefore x = c \frac{(p^2 + 1)}{p} = cp + \frac{c}{p}$

dist PG = $\sqrt{(cp + \frac{c}{p} - cp)^2 + (cp + \frac{c}{p} - \frac{c}{p})^2}$

$= \sqrt{(\frac{c}{p})^2 + (cp)^2}$

$= c\sqrt{p^2 + \frac{1}{p^2}}$

find min value of $p^2 + \frac{1}{p^2}$

Let $y = p^2 + \frac{1}{p^2} \quad y' = 2p - \frac{2}{p^3}$
 $= 0$ when $p = \frac{1}{p^3} \therefore p^4 = 1$

$\therefore p = \pm 1$

when $p = \pm 1 \quad y = 1 + 1 = 2$

But $p \neq \pm 1 \therefore y > 2$

Hence distance $> c\sqrt{2}$

[It is interesting to note that when $p=1$ Then $P=(c, c)$
 $\therefore P$ lies on the line $y=x$
 $\therefore y=x$ is the normal]

(b) (i) $\angle A = \angle D$ angles in same segm

$\therefore \frac{a}{\sin A} = \frac{a}{\sin D} = \frac{2r}{\sin \angle DBC}$

using sine rule in ΔBCD

$\angle DBC = 90^\circ$ (L in semicircle)

$\therefore \frac{a}{\sin A} = \frac{2r}{\sin 90^\circ} = 2r$

$\therefore \frac{a}{\sin A} = \frac{2r}{\sin 90^\circ} = 2r$



(ii)

Area ΔABC

$= \frac{1}{2} r^2 \sin \angle BOC + \frac{1}{2} r^2 \sin \angle AOC$

$+ \frac{1}{2} r^2 \sin \angle AOB$

But $\angle BOC = 2\angle A$ (Lat centre = 2 Lat circum)

etc

$\therefore \Delta ABC = \frac{1}{2} r^2 (\sin 2A + \sin 2B + \sin 2C)$